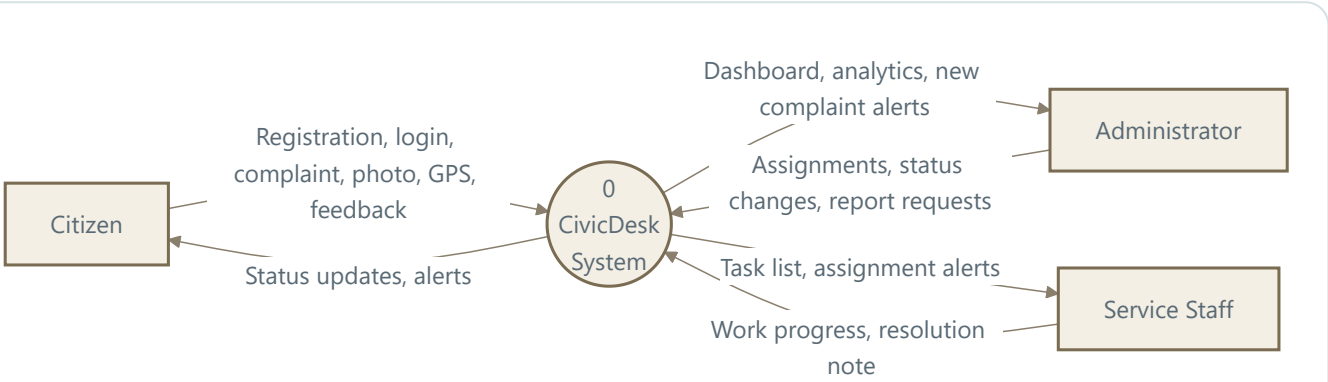
CivicDesk – Data Flow and UML Diagrams

Smart Community Service Management System. One data flow diagram (Levels 0, 1 and 2) and four UML diagrams: use case, class, sequence and state machine.

DFD 0 | DFD 1 | DFD 2 (2.0) | DFD 2 (3.0) | Use case | Class | Sequence | State

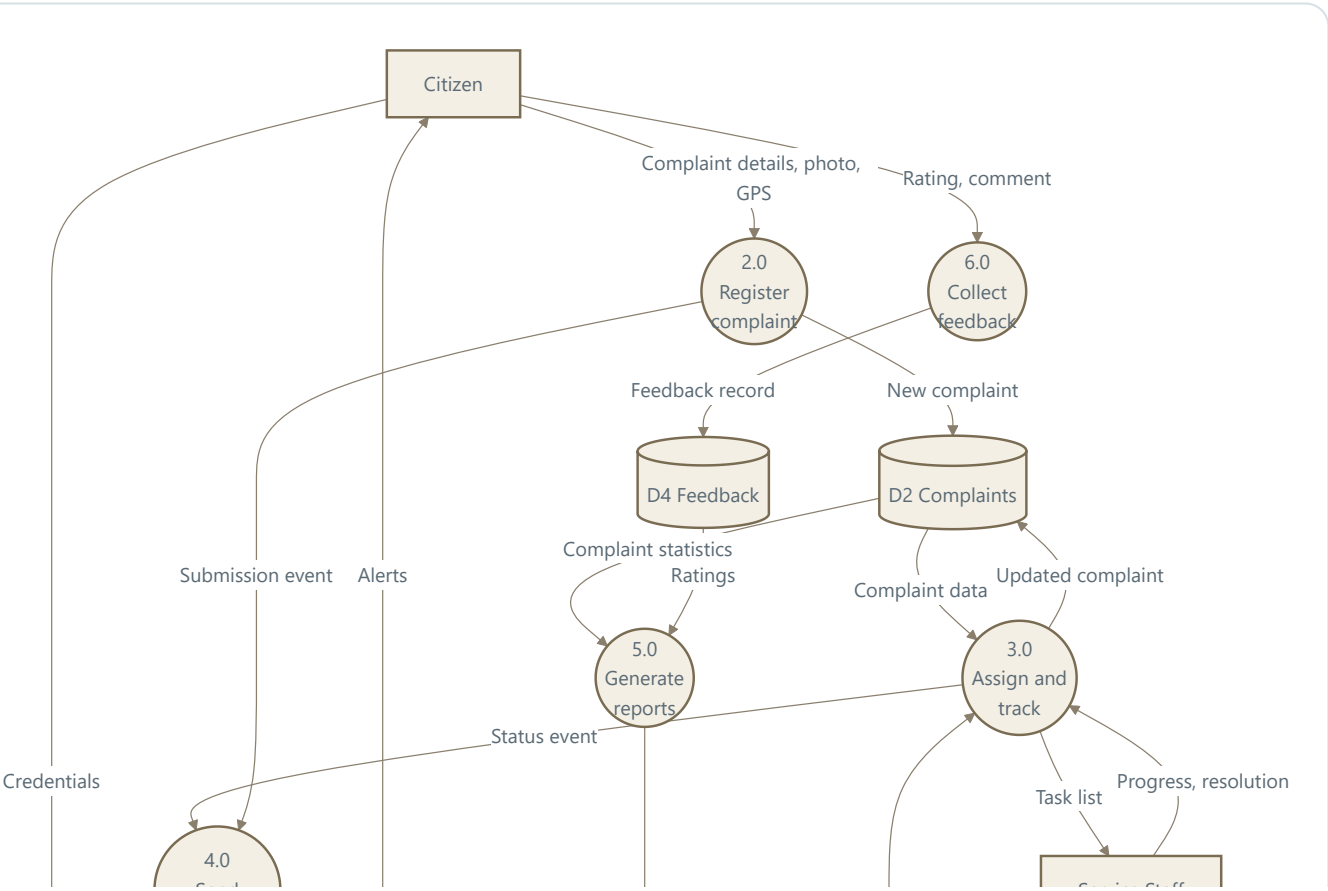
1. Data flow diagram, Level 0 (context)

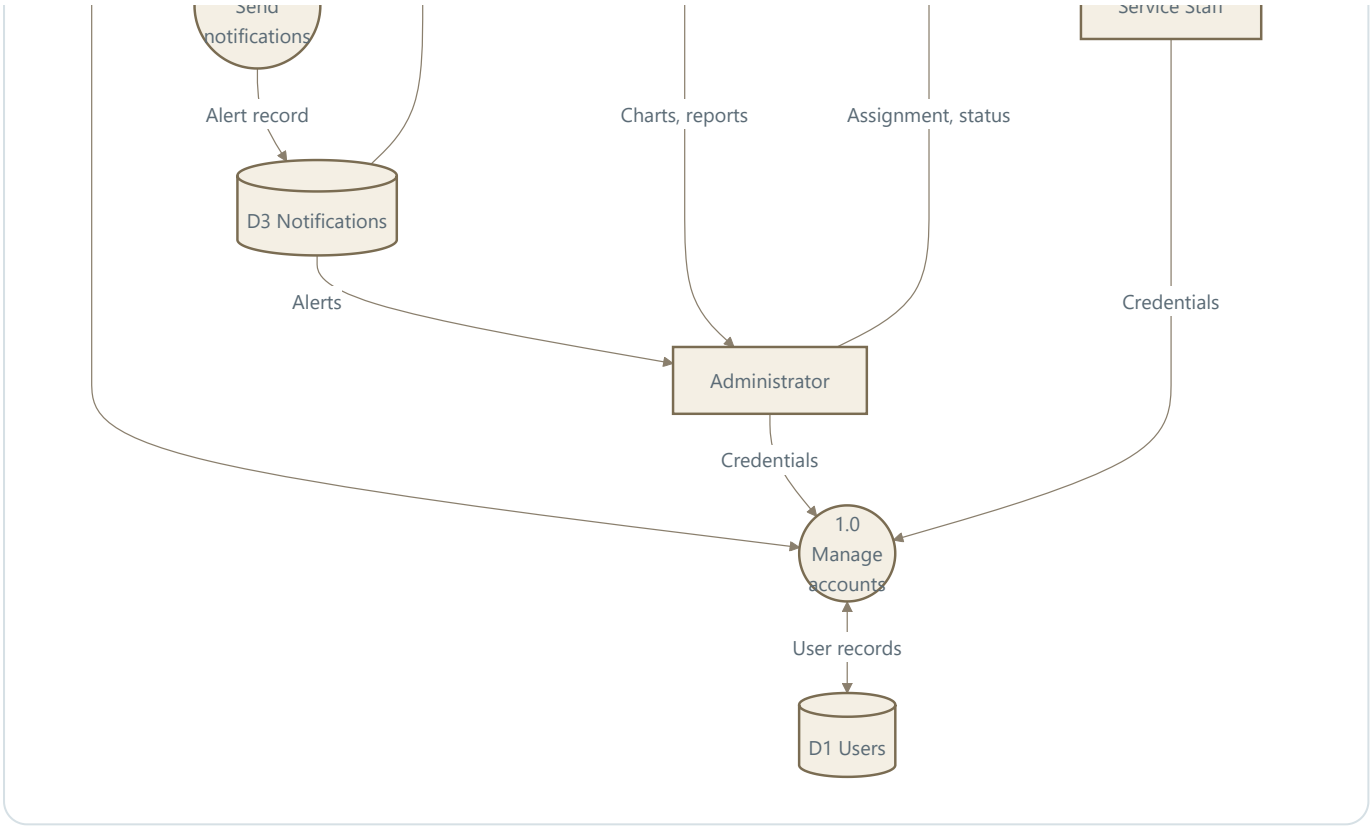
The whole system as one process, with the three external entities that exchange data with it.



2. Data flow diagram, Level 1

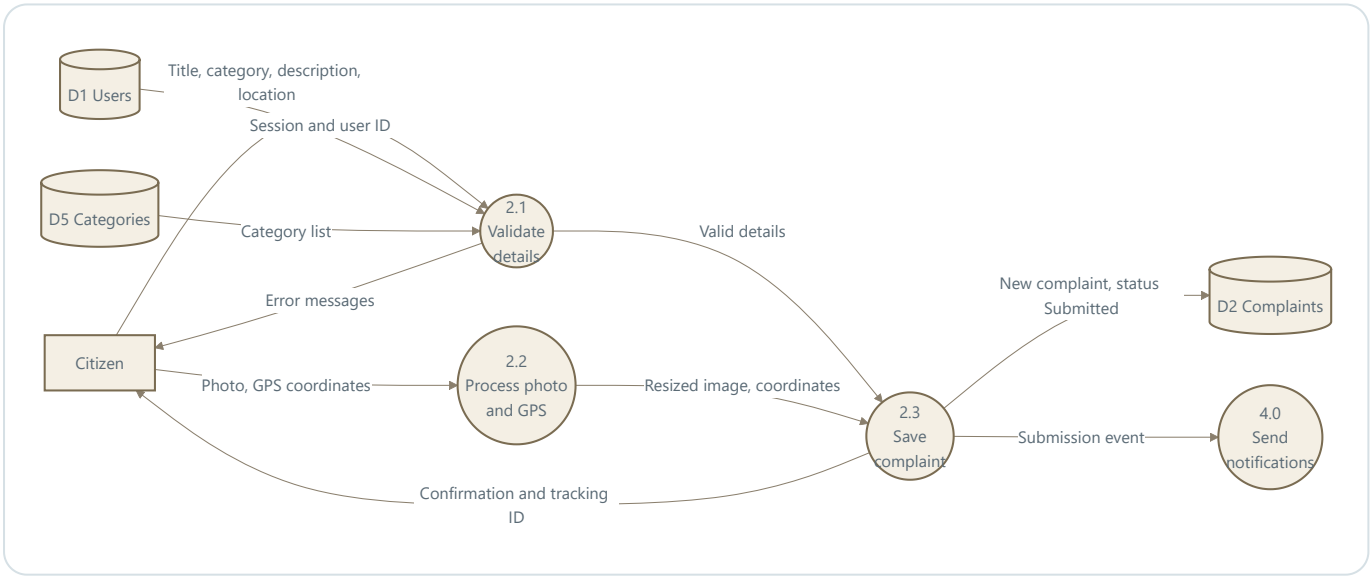
The system split into six processes and four data stores.





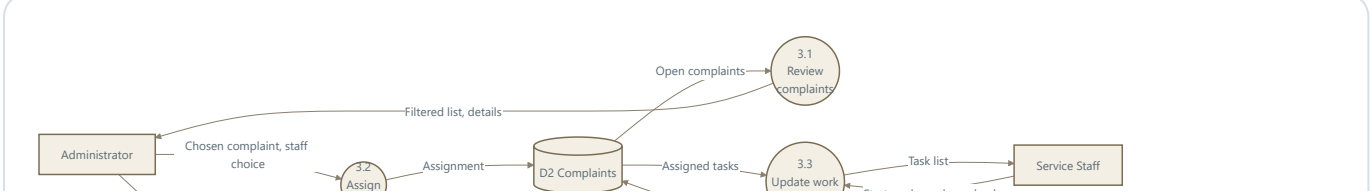
3. Data flow diagram, Level 2: process 2.0 Register complaint

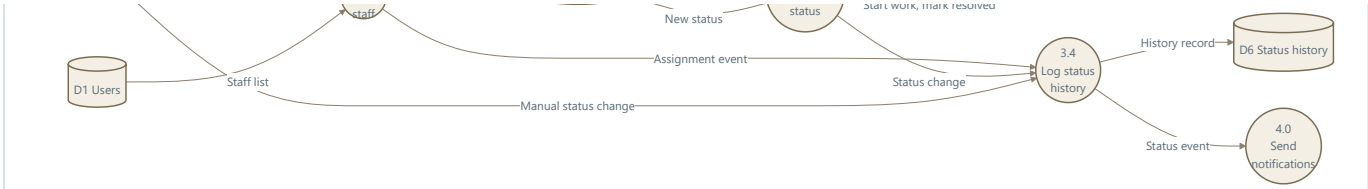
Process 2.0 from Level 1, broken into three sub-processes. Inputs and outputs match the Level 1 flows.



4. Data flow diagram, Level 2: process 3.0 Assign and track

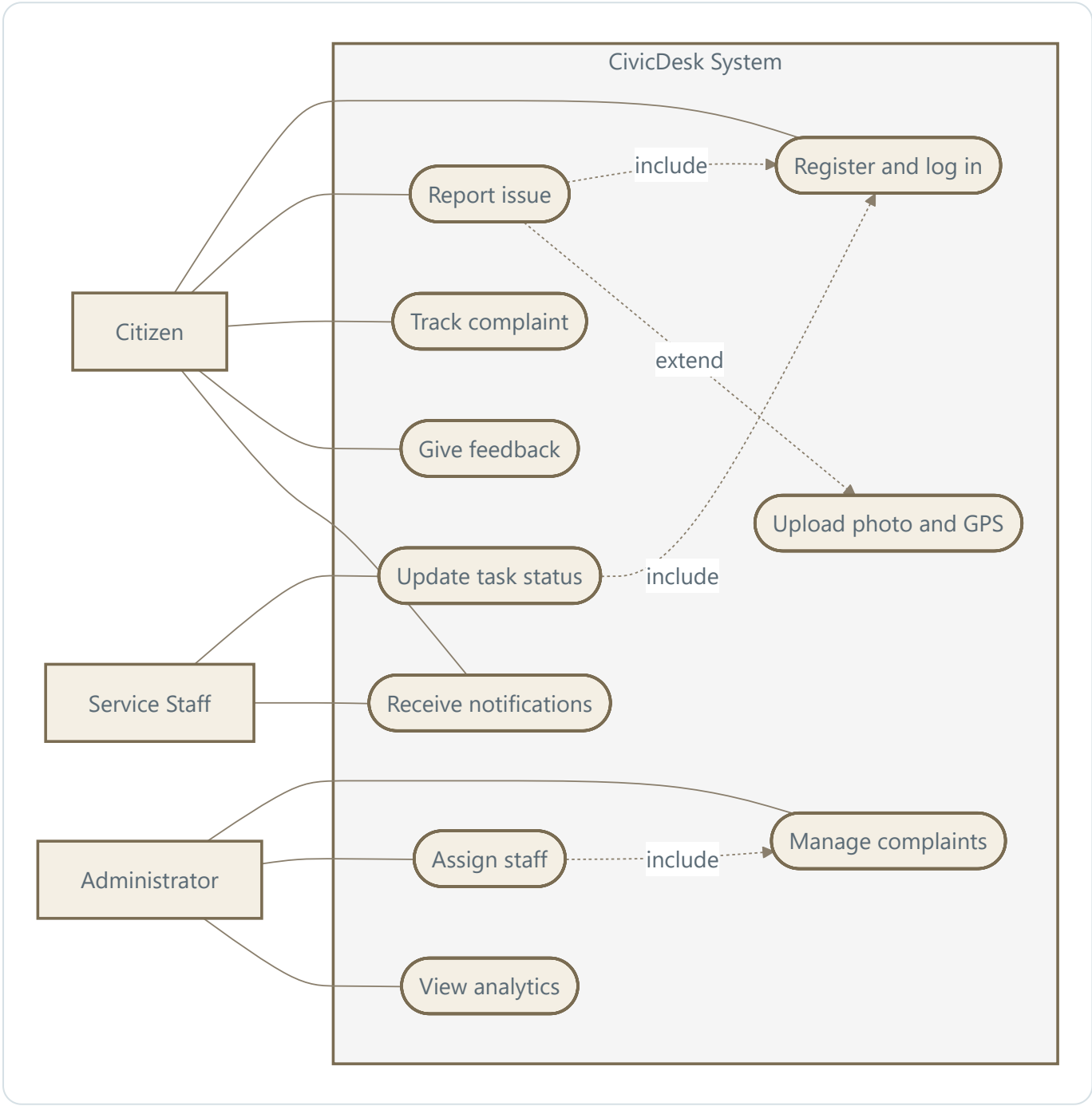
Process 3.0 from Level 1, broken into four sub-processes covering review, assignment, staff work and history logging.





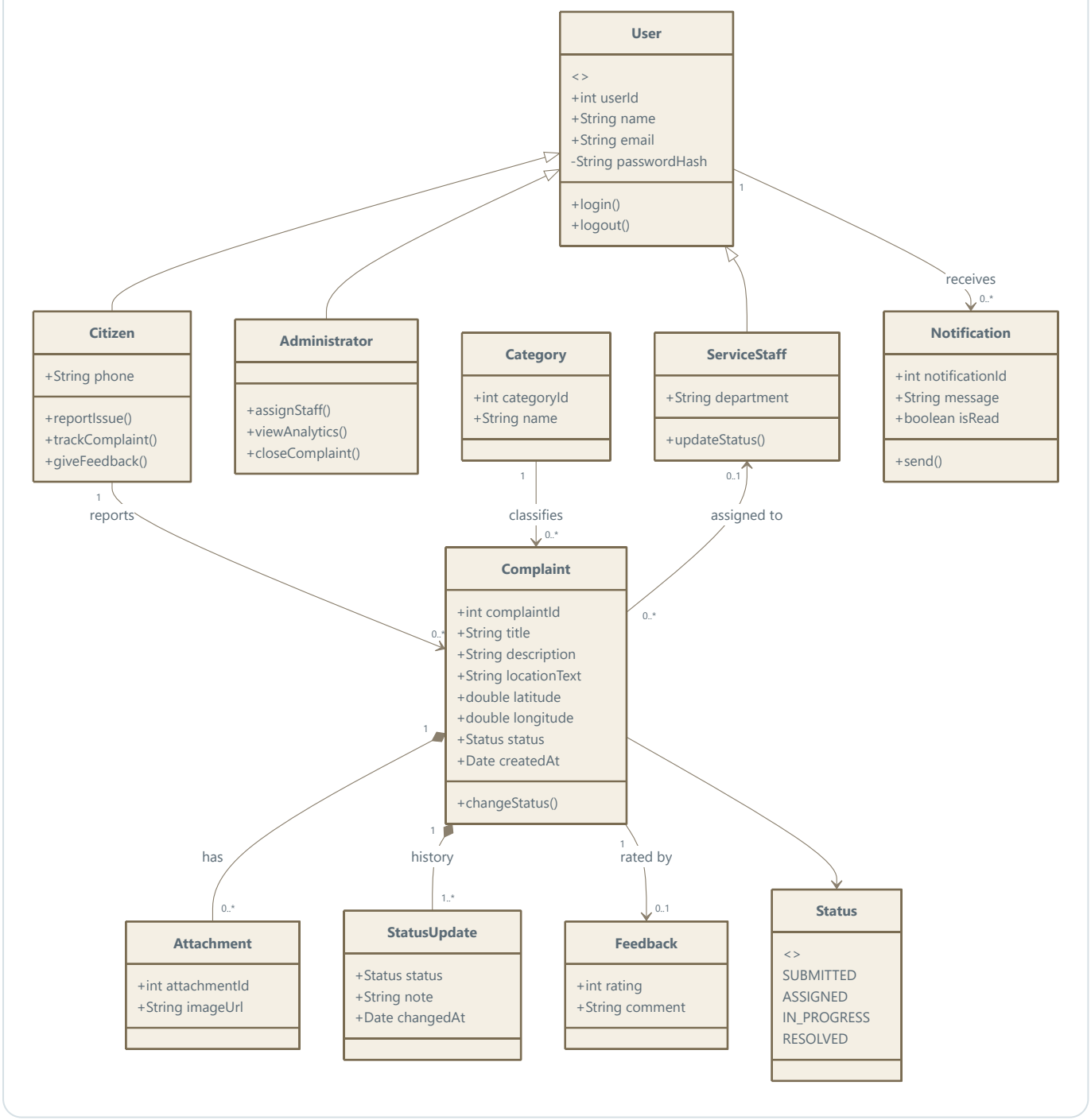
5. UML use case diagram

Who does what. Citizen, administrator and staff are the actors, and shared steps such as login are drawn as includes.



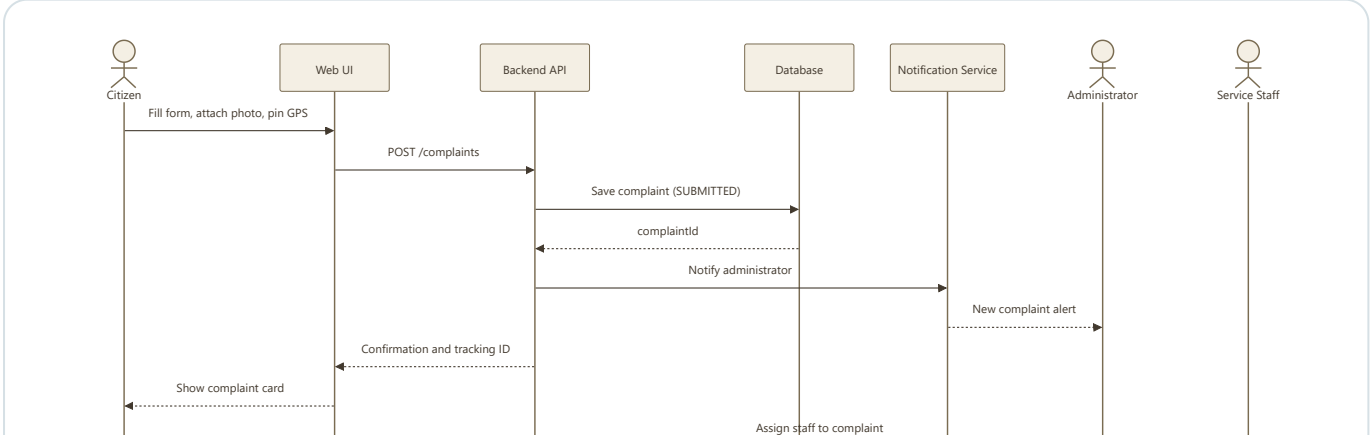
6. UML class diagram

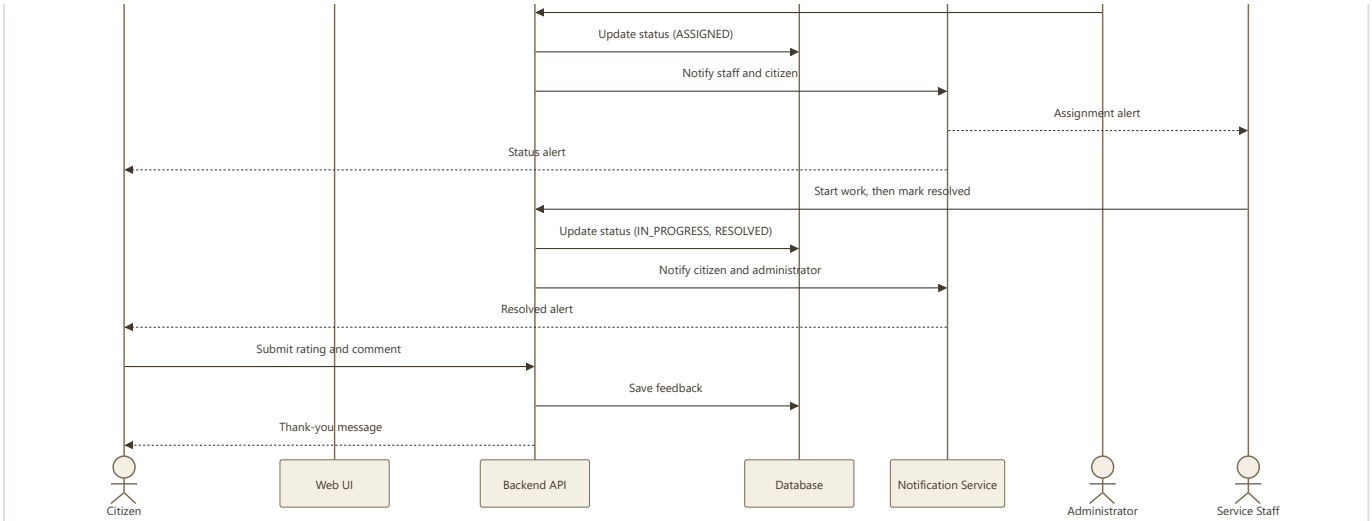
The main domain classes and how they relate.



7. UML sequence diagram

The main scenario: a citizen reports an issue, the administrator assigns it, staff resolve it, and the citizen rates the fix.





8. UML state machine diagram

The lifecycle of a single complaint, including a reopen path when the citizen is not satisfied.

